

SQL Date and Numeric Filtering Lab

Investigating Security Events Using SQL Date, Time, and Numeric Filters

In this lab, I applied advanced SQL filtering techniques within the MariaDB environment to investigate authentication events related to a simulated security incident. The investigation focused on filtering login attempts using dates, times, and event identifiers to isolate relevant records and improve the efficiency of incident analysis.

As a cybersecurity analyst, the ability to filter large datasets is critical when investigating suspicious activity. Security logs often contain thousands of events, making it necessary to narrow searches using operators such as '>', '<', '>=', '<=', and 'BETWEEN'.

This project demonstrates practical experience using SQL operators to investigate authentication activity and identify records relevant to a security incident.

Objectives

- Retrieve login attempts occurring after a specific date
- Filter authentication records within a defined date range
- Investigate login activity occurring outside normal business hours
- Analyze security events using numeric event identifiers
- Apply SQL comparison operators to security investigations
- Improve incident response and log analysis skills

Environment

Component	Details
Database	MariaDB
Database Name	Organization
Table Used	log_in_attempts
Operating Environment	Linux Bash Shell
Investigation Type	Authentication Log Analysis

Task 1 – Retrieve Login Attempts After a Certain Date

Query Login Attempts After May 9, 2022

SQL Query

```
SELECT *  
FROM log_in_attempts  
WHERE login_date > '2022-05-09';
```

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date > '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
35	tshah	2022-05-10	15:26:08	MEX	192.168.92.147	0
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
40	aalonso	2022-05-12	15:15:46	MEX	192.168.174.186	0
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
45	dtanaka	2022-05-11	10:28:54	US	192.168.223.157	1
46	eraab	2022-05-11	11:29:27	CAN	192.168.24.12	0
48	asundara	2022-05-11	03:18:45	USA	192.168.72.10	1
50	jclark	2022-05-10	10:48:02	CANADA	192.168.174.117	0
51	jrafael	2022-05-10	22:40:01	CANADA	192.168.148.115	1

Purpose

The purpose of this query was to identify all authentication events that occurred after May 9, 2022. This allowed the investigation to focus on activity occurring after the suspected incident date.

Query Login Attempts On or After May 9, 2022

SQL Query

```
SELECT *
FROM log_in_attempts
WHERE login_date >= '2022-05-09';
```

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date >= '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0

Purpose

This query expanded the investigation scope by including events that occurred on the incident date itself.

Analysis

The initial query identified **125 login attempts** that occurred after **2022-05-09**.

Narrowing authentication records to activity occurring after a specific date is a common incident response technique used to focus investigations on events that occurred after a suspected compromise or security alert.

When the scope was expanded to include **2022-05-09**, the result set increased to **165 login attempts**. This demonstrated how including a single day can significantly affect the volume of records available for analysis and emphasizes the importance of defining investigation timelines accurately.

Key Findings

- 125 login attempts occurred after 2022-05-09.
- 165 login attempts occurred on or after 2022-05-09.
- Date filtering effectively reduced the investigation scope.

Task 2 – Retrieve Logins Within a Date Range

Filter Login Attempts Between Two Dates

SQL Query

```
SELECT *
FROM log_in_attempts
WHERE login_date BETWEEN '2022-05-09' AND '2022-05-11';
```

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date BETWEEN '2022-05-09' AND '2022-05-11';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0

Purpose

The objective of this query was to limit the investigation to a specific timeframe surrounding the security incident.

Analysis

Filtering authentication records between **2022-05-09** and **2022-05-11** returned **123 login attempts**. Restricting the investigation to a defined timeframe provided a focused view of authentication activity surrounding the incident period.

Reviewing a smaller set of records improves investigation efficiency and reduces the likelihood of overlooking relevant events hidden within larger datasets.

Key Findings

- 123 authentication events occurred during the investigation period.
- Date range filtering reduced unnecessary records.
- Focused datasets improve incident analysis efficiency.

Task 3 – Investigate Logins at Certain Times

Retrieve Login Attempts Before Normal Business Hours

SQL Query

```
SELECT *  
FROM log_in_attempts  
WHERE login_time < '07:00:00';
```

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time < '07:00:00';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
27	aalonso	2022-05-10	01:55:35	MEX	192.168.103.210	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
48	asundara	2022-05-11	03:18:45	USA	192.168.72.10	1
55	jlansky	2022-05-11	05:15:34	US	192.168.6.170	0
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
59	rjensen	2022-05-12	04:52:08	MEX	192.168.54.140	0
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
75	zbernal	2022-05-12	04:14:35	US	192.168.188.63	1
78	smartell	2022-05-10	05:55:53	MEX	192.168.41.88	0

Purpose

The purpose of this query was to identify users accessing systems before normal working hours.

Retrieve Login Attempts Between 06:00 and 07:00

SQL Query

```
SELECT *  
FROM log_in_attempts  
WHERE login_time BETWEEN '06:00:00' AND '07:00:00';
```

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time BETWEEN '06:00:00' AND '07:00:00';
```

event_id	username	login_date	login_time	country	ip_address	success
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
98	gesparza	2022-05-11	06:30:14	CANADA	192.168.148.80	0
106	tmitchel	2022-05-12	06:15:41	MEXICO	192.168.3.252	1
134	iuduike	2022-05-09	06:46:40	USA	192.168.22.115	1
136	mabadi	2022-05-10	06:56:44	US	192.168.214.234	1
142	gesparza	2022-05-11	06:31:14	CANADA	192.168.117.56	1
147	yappiah	2022-05-08	06:04:34	MEX	192.168.65.245	0
148	daquino	2022-05-08	06:15:55	CANADA	192.168.135.6	1
182	lyamamot	2022-05-10	06:01:31	USA	192.168.106.52	0
191	cjackson	2022-05-08	06:46:07	CANADA	192.168.7.187	0
195	alevitsk	2022-05-11	06:59:13	CANADA	192.168.236.78	1

```
15 rows in set (0.001 sec)
```

Purpose

This query narrowed the investigation to a specific one-hour window immediately before business operations began.

Analysis

The investigation focused on authentication activity occurring outside standard business hours. Login attempts between **06:00:00** and **07:00:00** were isolated to identify potentially unusual user behavior before the start of the normal workday.

The earliest login observed within this timeframe occurred at **06:01:31**. Examining authentication events during off-hours can help identify compromised accounts, unauthorized access attempts, or employees accessing systems outside normal operating procedures.

Key Findings

- Early morning authentication activity was successfully isolated.
- The earliest login attempt occurred at **06:01:31**.
- Time-based filtering is valuable for identifying potentially suspicious activity.

Task 4 – Investigate Login Attempts by Event ID

Retrieve Events with IDs Greater Than or Equal to 100

SQL Query

```
SELECT event_id, username, login_date  
FROM log_in_attempts  
WHERE event_id >= 100;
```

```
MariaDB [organization]> SELECT event_id, username, login_date  
-> FROM log_in_attempts  
-> WHERE event_id >= 100;
```

event_id	username	login_date
100	tmitchel	2022-05-12
101	sbaelish	2022-05-08
102	jreckley	2022-05-09
103	jhill	2022-05-11
104	asundara	2022-05-11
105	cjackson	2022-05-12
106	tmitchel	2022-05-12
107	bisles	2022-05-12
108	daquino	2022-05-09
109	mcouliba	2022-05-10
110	mabadi	2022-05-09
111	aestrada	2022-05-10
112	rjensen	2022-05-09
113	gesparza	2022-05-10
114	smartell	2022-05-10
115	ivelasco	2022-05-10
116	tmitchel	2022-05-10
117	bsand	2022-05-08
118	smartell	2022-05-12
119	tmitchel	2022-05-11
120	tmitchel	2022-05-09
121	btang	2022-05-10
122	yappiah	2022-05-11
123	bmoreno	2022-05-10
124	asundara	2022-05-12
125	bisles	2022-05-11
126	jrafael	2022-05-12
127	abellmas	2022-05-09
128	jclark	2022-05-09

Purpose

The objective was to investigate a subset of authentication events based on their event identifier values.

Retrieve Events with IDs Between 100 and 150

SQL Query

```
SELECT event_id, username, login_date  
FROM log_in_attempts  
WHERE event_id BETWEEN 100 AND 150;
```



```
MariaDB [organization]> SELECT event_id, username, login_date  
-> FROM log_in_attempts  
-> WHERE event_id BETWEEN 100 AND 150;
```

event_id	username	login_date
100	tmitchel	2022-05-12
101	sbaelish	2022-05-08
102	jreckley	2022-05-09
103	jhill	2022-05-11
104	asundara	2022-05-11
105	cjackson	2022-05-12
106	tmitchel	2022-05-12
107	bisles	2022-05-12
108	daquino	2022-05-09
109	mcouliba	2022-05-10
110	mabadi	2022-05-09
111	astrada	2022-05-10
112	rjensen	2022-05-09
113	gesparza	2022-05-10
114	smartell	2022-05-10
115	ivelasco	2022-05-10
116	tmitchel	2022-05-10
117	bsand	2022-05-08
118	smartell	2022-05-12
119	tmitchel	2022-05-11
120	tmitchel	2022-05-09
121	btang	2022-05-10
122	yappiah	2022-05-11
123	bmoreno	2022-05-10
124	asundara	2022-05-12

Purpose

This query further narrowed the event scope to a smaller range of security records.

Analysis

Event identifiers were used to narrow the investigation to a targeted subset of authentication records. Filtering by event ID enabled efficient analysis of a specific range of security events without reviewing the entire log dataset.

Within the filtered results:

- The third returned record was associated with the login date **2022-05-09**.
- The fifth returned record belonged to the user **eraab**.
- The seventh returned record belonged to the user **tmitchel**.

These findings demonstrate how event identifiers can be used to quickly locate and review specific authentication events during security investigations.

Key Findings

- Event ID filtering reduced the volume of authentication records requiring review.
- The filtered dataset contained users from multiple departments and locations.
- Event-based filtering improved investigation precision and efficiency.

Investigation Summary

The SQL investigation demonstrated how date, time, and numeric filters can be used to efficiently analyze authentication activity during a security incident.

The investigation revealed:

- 125 login attempts occurred after the incident date.
- 165 login attempts occurred on or after the incident date.
- 123 authentication events occurred within the defined investigation window.
- Early-morning login activity was successfully isolated for review.
- The earliest observed login during the specified timeframe occurred at **06:01:31**.
- Event ID filtering enabled rapid identification of specific authentication records and users.

These techniques mirror real-world security operations workflows where analysts must quickly reduce large datasets into manageable and relevant records for investigation

What I Learned from This Lab

This lab strengthened my ability to use SQL as a tool for cybersecurity investigations. I gained practical experience filtering authentication logs using dates, times, and numeric identifiers to locate relevant security events. I also developed a better understanding of how security analysts reduce large datasets into manageable records during incident response, threat hunting, and security monitoring activities.

Key skills developed include:

- Date-based security log analysis
- Time-based authentication investigation
- Numeric filtering using event identifiers
- SQL comparison operators (**>**, **<**, **>=**, **<=**)

- Range filtering using **BETWEEN**
- Security event investigation and log analysis
- Database querying within a MariaDB environment

These skills are directly applicable to Security Operations Center (SOC) environments, incident response investigations, threat hunting, and security monitoring activities.

Key Takeaways

- Applied comparison operators (**>**, **<**, **>=**) to investigate authentication activity.
- Used **BETWEEN** to isolate records within defined date and event ranges.
- Analyzed login activity occurring outside normal business hours.
- Investigated authentication records using event identifiers.
- Improved proficiency in SQL-based security investigations and log analysis.
- Strengthened practical skills applicable to incident response, threat hunting, and SOC operations.

This version reads much more like a professional security investigation report and less like a lab worksheet.

Conclusion

This lab provided hands-on experience applying SQL operators to investigate authentication activity within a database environment. By filtering records using dates, times, and numeric identifiers, I was able to efficiently isolate relevant events and reduce investigation scope.

These SQL techniques are foundational skills for cybersecurity professionals involved in security monitoring, threat hunting, digital forensics, and incident response. The ability to accurately retrieve and analyze security-related data is critical for identifying suspicious activity and supporting effective security operations.

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